

WEST ORCA BAY BUOY 46060, AK, USA

WMO: 992850

Lat: 60.587N

Lon: 146.819W

Elev: 0

StdP: 14.70

Time Zone: -9.00 (NAK)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.2	23.4	N/A	N/A	N/A	N/A	N/A	N/A	42.4	38.1	38.9	38.3	13.2	20	0.762

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.6	62.5	N/A	61.0	N/A	59.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.4	32.5	29.2	DB	18.4	64.6	4.9	2.9	14.9	66.7	12.0	68.4	9.2	70.0	5.7	72.1	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	35.1	34.5	34.8	39.1	46.1	52.3	56.6	56.2	51.6	45.2	38.5	36.7		
	DBStd	9.07	5.45	4.66	4.25	3.53	3.77	3.41	2.52	2.34	2.83	3.64	5.15	4.52		
	HDD50	2760	462	433	470	328	129	14	0	0	16	151	344	413		
	HDD65	7683	927	853	935	778	586	382	260	273	403	614	794	878		
	CDD50	552	0	0	0	0	8	82	205	192	63	2	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	12.2	13.9	13.9	12.4	10.9	9.0	8.7	8.9	10.0	13.4	14.6	14.9	15.7		
(15) Precipitation	PrecAvg	56.1	5.4	4.7	3.8	2.9	2.9	2.8	3.8	5.2	7.0	6.1	5.4	6.1		
	PrecMax	72.1	12.8	10.1	9.6	5.6	5.9	4.5	6.3	9.5	11.9	12.3	12.5	12.0		
	PrecMin	38.8	1.2	0.1	1.1	0.9	0.8	0.8	2.0	1.1	2.8	3.0	0.1	2.0		
	PrecStd	7.0	2.4	2.1	2.1	1.1	1.4	0.9	1.0	2.2	2.4	2.4	2.9	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	44.1	43.2	51.1	57.5	64.4	65.8	64.4	58.8	53.4	49.0	46.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	43.6	42.3	41.9	47.6	54.4	60.7	62.9	62.2	57.3	51.7	47.6	44.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	42.4	41.2	41.0	45.2	52.5	58.7	61.1	60.9	56.4	50.7	45.9	43.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	41.4	40.1	40.1	43.3	51.1	56.8	60.0	59.8	55.4	49.7	44.7	42.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.2	4.7	4.5	4.3	3.9	3.6	3.5	3.7	3.7	4.5	4.5		
		MCDBR	4.7	4.3	5.1	6.1	6.5	6.4	5.3	5.4	4.2	4.1	4.9	4.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.311	0.323	0.363	0.382	0.370	0.384	0.377	0.344	0.323	0.274	0.403		
	taud		2.154	2.276	2.331	2.266	2.288	2.412	2.410	2.421	2.487	2.362	2.149	2.624		
	Ebn at Noon		151	214	255	263	266	273	265	257	248	208	143	97		
	Edh at Noon		13	21	27	35	37	33	33	30	24	19	12	9		
(44) All-Sky Solar Radiation	RadAvg		78	259	689	1186	1557	1645	1413	1081	718	323	111	44		
	RadStd		11	59	110	141	174	133	161	167	137	47	28	6		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (23 neighbors)		N/A	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page